PRASUN KUMAR

AI/ML Engineering | GenAI & Agentic AI | Solution Architecture & Technical Leadership

Senior Application Developer, IBM India

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PROFESSIONAL SUMMARY

AI/ML engineer with 9+ years of software-engineering experience and prior technical leadership at Harbinger Group. Builds enterprise GenAI, RAG, agent workflows, document intelligence and NLP/ML solutions using Python, Azure OpenAI and cloud-native services. Combines hands-on engineering with architecture reviews, mentoring, delivery planning and stakeholder coordination. Focuses on evaluation, human review, reliability, security and production readiness. Currently Senior Application Developer at IBM India, having joined on 3 November 2026.

CORE EXPERTISE

GenAI and agents: LangGraph, LangChain, Azure OpenAI, Amazon Bedrock, Llama, Qwen, Gemma, tool calling, state, memory, structured outputs and human approval.

RAG and document AI: Azure AI Search, embeddings, BM25/hybrid retrieval, reranking, FAISS, Chroma, OCR, Azure AI Document Intelligence and retrieval evaluation.

ML and LLMOps: scikit-learn, pandas, NumPy, PyTorch, TensorFlow, Hugging Face, NLP classification, LoRA/QLoRA, Databricks, MLflow and model evaluation.

Engineering and delivery: Python, FastAPI, SQL, PostgreSQL, MySQL, REST APIs, Azure Functions, Service Bus, Blob Storage, Docker, CI/CD, architecture reviews and mentoring.

PROFESSIONAL EXPERIENCE

Senior Application Developer, Band 07

IBM India Pvt. Ltd., Pune | Nov 2026 - Present

- Joined IBM India on 3 November 2026.

Associate AI Specialist (G3A)

Harbinger Group, Pune | Jul 2026 - Nov 2026

- Contributed to GenAI solution design, enterprise RAG, AI-powered automation testing, document workflows and LLM evaluation.

Associate Technical Lead

Harbinger Group | Jul 2024 - Jun 2026

- Led architecture and design reviews, code quality, mentoring, delivery planning and stakeholder coordination for AI/ML and GenAI initiatives.
- Designed document-processing and workflow solutions using Azure OpenAI, Azure AI Document Intelligence, Azure Functions and Service Bus, with validation, retries and review gates.

Senior Software Engineer

Harbinger Group | Apr 2021 - Jun 2024

- Developed Python services, machine-learning/NLP components, intelligent automation and enterprise integrations; collaborated with product, QA and engineering teams.

Software Professional

Extentia Information Technology | Nov 2019 - Apr 2021

- Built applied NLP prototypes for conversational assistants, resume parsing and semantic skill matching, alongside early computer-vision experiments.

Associate IT Applications Specialist / Master Level Intern

Symantec Software India Pvt. Ltd. | Jan 2017 - Oct 2019

- Built Python automation and application-support foundations. Full-time Associate IT Applications Specialist from July 2017; earlier period was an internship.

SELECTED PROFESSIONAL PROJECTS

Prior-role enterprise work. Public links are sanitized reference implementations, not employer source code.

GenAI Email-to-Case and Document Intelligence

- Designed extraction of email bodies and attachments into structured case records using Azure OpenAI, OCR/Azure AI Document Intelligence, Functions, Service Bus and Blob Storage.
- Applied schema validation, confidence scoring, idempotency, retries, audit trails and human review. Prior project results included a 60%+ improvement in extraction accuracy.

Public reference: email-to-case-genai-automation

Enterprise RAG and Requirements Discovery Agent

- Built document-ingestion, chunking, metadata-enrichment and retrieval approaches using Azure AI Search, embeddings, semantic search and keyword/hybrid retrieval.
- Used reranking, citation support and answer-quality evaluation to improve relevance and reduce unsupported answers; designed confidence-based review and no-answer handling.

Public reference: enterprise-rag-quality-lab

Agentic AI and Multi-Agent Workflow Automation

- Designed LangGraph/LangChain workflows with planner, retriever, validator and executor stages for multi-step enterprise tasks.
- Applied tool routing, state and memory management, approval gates, retries and auditable execution to control automated actions and support human oversight.

Public reference: agentic-ai-langgraph-workflows

AI-Powered Test Automation and Gap Assessment

- Worked on LLM-assisted test generation, document-to-test workflows and environment-aware Azure/AWS AI execution using Amazon Bedrock and evaluation tooling.
- Explored few-shot prompting, structured outputs, model comparisons and MLflow-tracked quality metrics with Databricks to support reviewable testing workflows.

Public reference: ai-test-automation-copilot

Behavioral Intelligence and ML/LLMOps

- Worked on behavior scoring and NLP classification, open-model evaluation with Qwen, Llama and Gemma, and parameter-efficient fine-tuning/adapter workflows.
- Applied threshold calibration, experiment tracking and Databricks evaluation to support model-quality assessment and reproducible comparisons.

INDEPENDENT GITHUB PROJECTS

Personal portfolio implementations, separate from professional client delivery.

ML Release Governance - Model training and paired holdout evaluation with bootstrap uncertainty, slice/drift checks and latency/cost release gates; tested evidence reports for human review.

AI Delivery Planner - Mixed-integer optimization of AI initiatives under budget, capacity, dependency and governance constraints; scenario comparison using illustrative planning inputs.

EDUCATION AND RECOGNITION

MCA - Savitribai Phule Pune University, 2017 | BCA - Sikkim Manipal University, 2014.

Post Graduate Program in Machine Learning and Artificial Intelligence.

ML and AI credential verification | Technical and team awards; hackathon recognition (2017).